

WORKSHOP #1 - ANALYSIS

Alejandro Ramirez Nuñez
Juan Felipe Quiroga Medina
Juan Camilo Lopez Leon
Juan David Yopasa Maldonado

2026-2S

Ingeniería de Software II
Liliana Marcela Olarte

Universidad Nacional de Colombia
Facultad de Ingeniería
Sede Bogotá

1. Requirements Documentation

Functional Requirements

ID	Name	Description	Priority	Dependency
FR_01	Interactive construction sites map	The system must display to any user, registered or not, a map of Bogotá representing each active construction site in its georeferenced location, using a marker for specific construction sites and a line path for linear construction sites.	High	None
FR_02	Map zoom	The system must allow any user, registered or not, to zoom in and out of the map.	High	FR_01
FR_03	Construction site name on map	The system must allow any user, registered or not, to view the name of the construction site by clicking on its location on the map.	High	FR_01
FR_04	Construction site detail sheet	The system must show any user, registered or not, a detail sheet for each construction site with name, type, locality, physical progress percentage, performance index, dates (start, contractual end, and estimated end), contractor, budget, and image gallery.	High	None
FR_05	Export construction site to PDF	The system must allow registered users to export the data of a construction site in PDF format.	Medium	FR_04, FR_16
FR_06	Construction site subscription	The system must allow registered users to subscribe to a construction site to receive periodic updates via email regarding physical progress or estimated completion date.	Low	FR_04, FR_16
FR_07	Construction site filtering	The system must allow any user, registered or not, to filter the search for construction sites by locality,	Medium	None

		physical progress percentage, type of construction site, and responsible contractor.		
FR_08	Construction site rating	The system must allow registered users to leave reviews from 0 to 5 stars indicating satisfaction with a particular construction site.	High	FR_04, FR_16
FR_09	Construction site comments	The system must allow registered users to leave comments associated with a particular construction site.	High	FR_04, FR_16
FR_10	Delete own comments	The system must allow registered users to delete the comments they have published.	High	FR_09
FR_11	Report comments	The system must allow registered users to report comments from other people.	Medium	FR_09, FR_16
FR_12	Comment moderation	The system must allow the administrator to delete any comment.	High	FR_09, FR_30
FR_13	Construction site sorting	The system must allow any user, registered or not, to sort construction sites by delay, budget, physical progress, start date, and estimated completion date.	Medium	None
FR_14	Petition template download	The system must allow registered users to download the right of petition template solely in PDF format.	High	FR_15, FR_16
FR_15	Pre-filled petition template	The system must provide the template pre-filled with generic data regarding the entity to which the request is directed, contact details of the requester, a clear description of what is requested, supporting reasons, relevant documents or evidence, and preferred response channel, allowing the user to replace them with their information before filing, in accordance with the requirements	High	None

		established by Law 1755 of 2015 [4] for filing a right of petition		
FR_16	User login	The system must allow registered users to log in once registered on the page.	High	FR_17
FR_17	User registration	The system must allow users to register using an email account.	High	None
FR_18	Password recovery	The system must allow registered users to recover their password by sending a message to their associated email.	High	FR_17
FR_19	Construction site search	The system must allow any user, registered or not, to perform a search returning matches in the name or description of the construction sites.	High	None
FR_20	Profile editing	The system must allow registered users to edit their profile data: profile photo, name, description, and password.	High	FR_16
FR_21	Comment history	The system must allow registered users to view their comment history upon entering their profile, displaying the content and publication date of each comment.	Medium	FR_09, FR_16
FR_22	Construction site comparison	The system must allow registered users to compare two construction sites, reflecting differences in physical progress, performance index, contractor, and budget.	Medium	FR_16
FR_23	Locality comparison	The system must allow registered users to compare two localities, showing average performance index, number of active construction sites, and contractors with the most construction sites in each locality.	Medium	None

FR_24	Contractor comparison	The system must allow registered users to compare two contractors, showing the number of construction sites and average performance index for each contractor.	Medium	None
FR_25	Administrator dashboard	The system must display a dashboard to the administrator with all active construction sites and their associated information.	High	FR_30
FR_26	Construction site management	The system must allow the administrator to manually add, edit, and delete construction sites and their associated information.	High	FR_30
FR_27	Construction site evolution graph	The system must display to any user, registered or not, an evolution graph of the construction site over time.	Medium	None
FR_28	Construction site milestone timeline	The system must display to any user, registered or not, a timeline showing key project milestones.	Medium	None
FR_29	Road closures	The system must show any user, registered or not, the road sections closed due to construction sites.	Medium	None
FR_30	Administrator login	The system must allow the administrator to log in with credentials assigned by the system, granting access to the administration functions.	High	None

Non-Functional Requirements

ID	Name	Description	Category	Priority	Dependencies
NFR_01	Construction site listing response time	The system must respond to construction site listing requests in under 5 seconds, returning a maximum of 20 records per page, for datasets of up to 200 stored construction sites.	Performance	High	FR_07, FR_13
NFR_02	Map marker clustering	The system must automatically cluster map icons when the user reduces the zoom level of the display.	Usability	Medium	FR_01, FR_02
NFR_03	Construction site query rate limiting	The system must limit the number of requests per user to 60 per minute when attempting to consult a specific construction site.	Reliability	Medium	FR_04
NFR_04	Comment posting limit	The system must limit comment posting by the same user to a maximum of 1 comment per minute, in order to prevent	Security	Medium	FR_09

		automated or repetitive posts.			
NFR_05	Secure error handling	The system must handle all unhandled backend errors without exposing internal information (file paths, stack traces, SQL queries) to the client, returning a generic error message and an internal reference code for debugging instead.	Security	High	None
NFR_06	Account enumeration prevention	Error messages visible to the user must not reveal whether an email or user exists in the system.	Security	Medium	FR_16, FR_17, FR_18
NFR_07	Construction site detail response time	Queries for information regarding a construction site must not take more than 2 seconds in 95% of requests.	Performance	High	FR_04
NFR_08	Search response time	The system must take a maximum of 5 seconds to return search results performed by a user in 95% of requests.	Performance	Medium	FR_19

NFR_09	Password hashing	The system must store user passwords using a hashing algorithm, ensuring passwords are not stored in plain text [5].	Security	High	FR_17, FR_20
NFR_10	Responsive interface	The system must adapt its interface for screen widths between 320 pixels and 1920 pixels without generating horizontal scrolling.	Compatibility	High	None
NFR_11	Construction sites volume scalability	The system must maintain the response times defined in NFR_01, NFR_07, and NFR_08 when the number of registered construction sites increases from 200 to 500.	Scalability	Low	NFR_01, NFR_07, NFR_08
NFR_12	Single-command deployment	The system must allow complete deployment of backend, frontend, and database through a single docker-compose command.	Maintainability	High	None

NFR_13	Personal data consent	The system must request and log explicit user consent for processing personal data before completing registration, in compliance with Law 1581 of 2012 [2].	Legal/Regulatory	High	FR_17
NFR_14	Data source attribution	The system must indicate the source and update date of data for each published construction site, referencing the origin, in accordance with the principles of public information access established in Law 1712 of 2014 [3]	Legal/Regulatory	Medium	FR_04
NFR_15	Administrator access restriction	The system must restrict access to administration functions (dashboard, construction site management, and comment moderation) exclusively to accounts with the administrator role, rejecting any request from other users.	Security	High	FR_12, FR_25, FR_26, FR_30

2. User Stories

US-01: Map Navigation

User Story

As a user, I want to navigate through the interactive map displaying all active construction sites in Bogotá D.C., so that I can visualize all the ongoing operations in a specific location.

Acceptance Criteria

Given that the user is navigating through the map to check active constructions, when the user focuses on a specific point of interest on the map, then the system shall display all the construction sites that are currently active in that location.

Related Requirements

FR_01, FR_02, FR_03, NFR_02

US-02: Consult information of a construction site

User Story

As a user, I want to visualize the information of a construction site when I click on an active construction site on the interactive map, so that I can check all the relevant information of that site (name, location, status and due date).

Acceptance Criteria

Given that the user wants to see more information on a specific construction site, when the user clicks on a specific construction site on the interactive map, then the system will show a data sheet of the site (name, exact location, status, progress and due date).

Related Requirements

FR_04, NFR_03, NFR_07, NFR_14

US-03: Export data sheet of a construction site

User Story

As a user, I want to save a data sheet of a specific construction site in PDF format, so that I can store relevant site information on my computer, phone, or tablet.

Acceptance criterion

Given that the user wants to save relevant construction site information to their device, when the user selects the "Export Data Sheet" option, then the system opens a window allowing the user to specify the file name and save location; upon clicking the Export button, the file will be downloaded to their device.

Related Requirements

FR_05

US-04: Alerts and notifications

User Story

As a user, I want to subscribe to a notification system, so that I receive updates on my email regarding progress and due date for selected construction site at regular intervals of time.

Acceptance criterion

Given that the user cannot frequently visit the website to stay updated on the progress of a construction site, when the user subscribes their email to receive notifications for a specific site, then the system will send periodic email containing all relevant information and updates.

Related Requirements

FR_06

US-05: Search Engine**User Story**

As a user, I need a filter system in the search engine, so that I can search for information by locality, physical progress percentage, type of construction site, and responsible contractor.

Acceptance criterion

Given that the user requires a search tailored under certain parameters, when selecting an option from the filters (locality, physical progress percentage, type of construction site, responsible contractor), then the system will search for matches in the specified fields of the selected filters.

Related Requirements

FR_07, NFR_01

US-06: Reviews**User Story**

As a user, I need to rate the progress and status of a construction site using a star rating system from 0 to 5, where 5 represents the maximum level of satisfaction and 0 the minimum, so that I can give my objective opinion about the status of a construction site in the city of Bogotá.

Acceptance criterion

Given that users want to give feedback through a rating system, when selecting a construction site rating score (ranging from 0 to 5 stars), then the system will display a pop-up with the message "Thank you for your feedback", and immediately update the construction site's average rating.

Related Requirements

FR_08

US-07: Comments**User Story**

As a user, I need to leave a comment on construction sites I choose, so that I can express my opinion about a construction site and inform other users about its status.

Acceptance criterion

Given that I wish to express my specific opinion regarding the status or progress of a construction site in Bogotá, when I create a comment and upload it to the system, then it will be posted on the construction site's detail page where other users can read it.

Related Requirements

FR_09, NFR_04

US-08: Rights of Petition**User Story**

As a user, I need a tool to help me draft a right of petition when I feel a construction site has been stalled for a long time or lacks information on district web pages, so that I can submit a right of petition more easily.

Acceptance criterion

Given that the user wishes to file a right of petition, when selecting the option "Submit my right of petition", then the page will direct them to a template with detailed instructions on how to present the petition and the steps to file it correctly.

Related Requirements

FR_14, FR_15

US-09: Login

User Story

As a user, I need to log into the page using my credentials (username and password), so that I can access features like rating construction sites, leaving comments, and accessing the Right of Petition tool.

Acceptance criterion

Given that the user wants to access services offered to registered users, when logging in using credentials, then the system unlocks registered user privileges and grants access to profile settings, comments, reviews, and created rights of petition.

Related Requirements

FR_16, NFR_06

US-10: Registration

User Story

As a user, I need to register on the platform to access services provided to registered users, so that I can log in later and save all my activity under the created profile.

Acceptance criterion

Given that the user wishes to create a profile on the platform, when filling out the registration form and clicking the Register button, then the system will display a pop-up saying "Registration successful", after which the user can log in with their username and password.

Related Requirements

FR_17, NFR_09, NFR_13

US-11: Comparison Chart

User Story

As a user, I need to compare two construction sites, localities, or contractors based on their features, so that I can form a critical view on project developments that seem similar.

Acceptance criterion

Given that the user is searching and wants to compare two similar items, when two construction sites, localities, or contractors are selected and the user clicks compare, then a comparison window will appear.

Example of comparison window:



Related Requirements

FR_22, FR_23, FR_24

US-12: Administrator Dashboard

User Story

As an administrator, I need a dashboard to review active construction site activity and associated information, so that I can effectively monitor all active projects along with their details.

Acceptance criterion

Given that the administrator logs into the page with admin privileges and wishes to view key site information efficiently, when logging in with credentials, then the system redirects to the admin dashboard displaying all active construction sites and their details.

Related Requirements

FR_25, FR_30, NFR_15

US-13: Add, edit, and delete construction sites

User Story

As an administrator, I need to add, edit, and delete construction sites, so that I can manage the construction construction sites listed on the platform.

Acceptance criterion

Since the administrator needs to add, edit, and delete a construction site on the page, selecting the pencil icon opens a window allowing them to choose from these three options. An example of the pencil icon is shown; it was chosen because a pencil is associated with creating, editing, and deleting.



Related Requirements

FR_26, NFR_15

US-14: Evolution of the construction site

User Story

As a user, I need to see the progress of the project so that I can track the changes and advancements over time.

Acceptance criterion

Since a user reviewing a specific project wants to see its progress over time, selecting the function icon causes the system to open a window displaying a graph that shows the project's progress over time.

Example of a function icon.



Related Requirements

FR_27

US-15: Delete own comments**User Story**

As a registered user, I want to delete the comments I have published, so that I can remove opinions that are no longer accurate or that I no longer wish to share.

Acceptance criteria

Given that the registered user is logged in and is viewing a comment they published, when the user selects the "Delete" option and confirms the action, then the system removes the comment from the construction site's detail page and from the user's comment history.

Given that the user is viewing a comment published by another user, when the comment is displayed, then the system does not show the "Delete" option for that comment.

Related Requirements

FR_10

US-16: Report comments**User Story**

As a registered user, I want to report comments from other users that I consider offensive, false, or inappropriate, so that the administrator can review them and keep the discussion respectful and reliable.

Acceptance criterion

Given that the registered user is logged in and is reading a comment published by another user, when the user selects the "Report" option and chooses a reason, then the system registers the report, displays the message "Report sent", and makes the comment available for review in the administration panel.

Related Requirements

FR_11

US-17: Comment moderation**User Story**

As an administrator, I want to delete any comment published on the platform, so that I can remove offensive, false, or spam content and keep the information reliable for citizens.

Acceptance criteria

Given that the administrator is logged in and is reviewing a reported comment, when the administrator selects "Delete" and confirms the action, then the system permanently removes the comment from the construction site's detail page.

Related Requirements

FR_12, FR_30, NFR_15

US-18: Sort construction sites**User Story**

As a user, I want to sort the list of construction sites by delay, budget, physical progress, start date, and estimated completion date, so that I can quickly identify the most delayed construction sites or the ones most relevant to me.

Acceptance criterion

Given that the user is viewing the list of construction sites, when the user selects a sorting criterion (delay, budget, physical progress, start date, or estimated completion date) and an order (ascending or descending), then the system reorders the list according to the selected criterion, showing a maximum of 20 construction sites per page.

Related Requirements

FR_13, NFR_01

US-19: Keyword search**User Story**

As a user, I want to search for construction sites by typing words from their name or description, so that I can find a specific construction site without having to locate it on the map.

Acceptance criteria

Given that the user is on the construction sites list or map page, when the user types a keyword in the search bar and confirms the search, then the system displays the construction sites whose name or description contain that keyword in no more than 5 seconds.

Given that no construction site matches the entered keyword, when the search is performed, then the system displays the message "No construction sites match your search".

Related Requirements

FR_19, NFR_08

US-20: Password recovery**User Story**

As a registered user, I want to recover my password through my registered email, so that I can regain access to my account if I forget it.

Acceptance criterion

Given that the user is on the login page and has forgotten the password, when the user clicks "Forgot password" and enters an email, then the system displays the message "If the email is registered, you will receive a recovery link" and sends a password reset link only if the email belongs to an existing account.

Related Requirements

FR_18, NFR_06

US-21: Profile editing**User Story**

As a registered user, I want to edit my profile data (profile photo, name, description, and password), so that my account information stays up to date.

Acceptance criteria

Given that the registered user is logged in and is on the profile page, when the user modifies one or more fields and clicks "Save changes", then the system updates the profile data and displays the message "Profile updated successfully".

Given that the user wants to change the password, when the user enters a new password, then the system requests the current password before saving the change.

Related Requirements

FR_20, NFR_09

US-22: Comment history**User Story**

As a registered user, I want to see the history of the comments I have published from my profile, so that I can keep track of my participation in the oversight of construction sites.

Acceptance criterion

Given that the registered user is logged in, when the user enters the profile page, then the system displays the list of published comments with their content, publication date, and associated construction site, ordered from newest to oldest.

Related Requirements

FR_21

US-23: Milestone timeline**User Story**

As a user, I want to see a timeline with the key milestones of a construction site, so that I can understand which stage the construction site is in and which milestones have been met or delayed.

Acceptance criterion

Given that the user is viewing the detail sheet of a construction site, when the user opens the "Timeline" section, then the system displays the key milestones of the construction site in chronological order with their planned dates, indicating which ones have been completed.

Related Requirements

FR_28

US-24: Road closures**User Story**

As a user, I want to see the road sections closed due to construction construction sites, so that I can plan my routes around the city and avoid affected areas.

Acceptance criterion

Given that the user is viewing a single construction site, when the user enables the "Road closures" layer, then the system highlights the road sections closed by construction construction site.

Related Requirements

FR_29

US-25: Administrator login**User Story**

As an administrator, I want to log in with credentials assigned by the system, so that I can access the administration functions of the platform.

Acceptance criteria

Given that the administrator is on the login page, when the administrator enters valid administrator credentials, then the system grants access to the administration functions and redirects to the administrator dashboard.

Given that a registered user without the administrator role is logged in, when the user tries to access an administration page, then the system denies access with an authorization error.

Related Requirements

FR_30, NFR_15

3. User Story Mapping

Visitor (no account)			Registered citizen				Administrator
Explore the map	Consult a work	Find works	Manage account	Give feedback	Take action	Compare	Administer
See works	Open work sheet	Search by keyword	Sign up	Rate a work	Draft a petition	Two works	Sign in
Move around	Check data source	Filter	Sign in	Comment	Export a work	Two localities	Update works
Identify a work	Follow progress	Sort	Recover access	Manage my comments	Follow updates	Two contractors	Monitor
			Edit profile				Moderate

Release 1 • MVP (walking skeleton)

US-01 Map navigation	US-02 Work information	US-19 Keyword search	US-10 Registration	US-07 Comments		US-25 Admin login
			US-09 Login			US-13 Manage works
						US-12 Admin dashboard
						US-17 Comment moderation

Non-functional: NFR_04, NFR_05, NFR_06, NFR_07, NFR_08, NFR_09, NFR_10, NFR_12, NFR_13, NFR_14, NFR_15

Release 2 • Participation and search

US-05 Search filters	US-20 Password recovery	US-06 Reviews	US-08 Right of petition	US-11 Comparison chart
US-18 Sort works	US-21 Profile editing	US-15 Delete own comments		

Non-functional: NFR_01

Release 3 • Insights and extras

US-24 Road closures	US-14 Work evolution	US-22 Comment history	US-16 Report comments	US-03 Export data sheet
	US-23 Milestone timeline			US-04 Alerts and notifications

Non-functional: NFR_02, NFR_03, NFR_11

Top rows: users, activities and user tasks (backbone). Cards: user stories (section 2, where each lists its related requirements). Blue: visitor · Green: registered citizen · Orange: administrator.

4. Use of Artificial Intelligence

This project used Claude (Anthropic) as a support tool during the requirements analysis phase. Its use was limited to the following tasks:

- **Review and refinement of functional and non-functional requirements:** verifying the completeness and correctness of the dependencies declared between requirements (ID column) and checking wording consistency across the requirement tables.
- **User Story Map schema generation:** producing the visual layout of the user story map (actors, activities, backbone tasks, release grouping, and cross-cutting non-functional requirements) based on the user stories and requirements already defined by the team.

5. References

1. Plan.io. *User Story Mapping*. Retrieved from <https://plan.io/blog/user-story-mapping/>
2. Congreso de la República de Colombia. (2012). *Ley 1581 de 2012, por la cual se dictan disposiciones generales para la protección de datos personales*.
3. Congreso de la República de Colombia. (2014). *Ley 1712 de 2014, Ley de Transparencia y del Derecho de Acceso a la Información Pública Nacional*.
4. Congreso de la República de Colombia. (2015). *Ley 1755 de 2015, por medio de la cual se regula el Derecho Fundamental de Petición*.
5. Django Software Foundation. *Django Documentation — Password Management*. Retrieved from <https://docs.djangoproject.com/en/stable/topics/auth/passwords/>